Cross-Linguistic Computational Analysis of Copula Deletion

A Cross-Register Investigation of Indo-Aryan & Dravidian Languages

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Course : Language , Cognition & Computation (HS5652)

Presentation Road Map:

Cross-Linguistic Computational Analysis of Copula Deletion

A Cross-Register Investigation of Indo-Aryan & Dravidian Languages

01 *Introduction*

Assignment Introduction & Overview

Presenter: **Sairam**

02 *Literature*

Literature Review & Theoretical Background

Presenter: **Varsha**

03 *Research Gap*

Research Gap & Approach to Gap

Presenter: **Arya**

04 *Computation*

Data Analysis, Extraction & Computational Methodology

Presenter: **Nitish**

05 *Discoveries*

Gaps, Novel Traits & Key Findings

Presenter: **Swapnil**

06 *Conclusion*

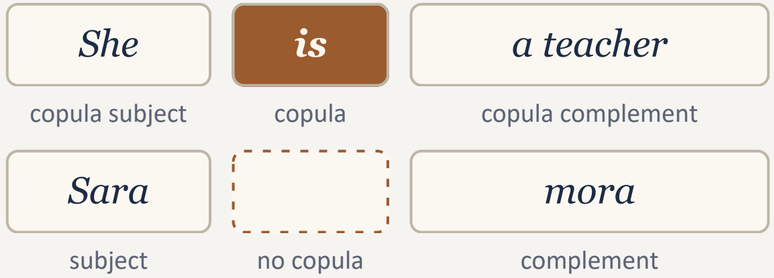
Future Scope & Concluding Remarks

Presenter: **Karthik**

What Is a Copula?

A **copula** is a linking word that joins a subject to a non-verbal predicate, like English “is” in “She is a teacher.” It adds little meaning of its own; it mainly carries tense, person and negation. In a **zero-copula** clause the subject and predicate simply sit side by side.

ANATOMY OF A COPULA CLAUSE



Hebrew: *Sara mora* ‘Sara (is a) teacher’

45% of 386 languages allow a zero copula with predicate nouns (Stassen, WALS ch. 120)

1 What a copula clause expresses

Meaning	English example	Our predicate tag
Identity	*“That man is my father.”*	**Equational / Nominal**
Class membership	*“She is a doctor.”*	**Equational / Nominal**
Property	*“The food is very good.”*	**Adjectival**
Location / existence	*“Mother is at home.”*	**Locative / Existential**

2 When is the copula zero?

Absent or dropped?
Some languages have no present copula for ‘X is Y’ at all (zero is the grammar); others have one that speakers may leave out (zero is a choice).

Only in some contexts
Zero often depends on tense, person and polarity: Russian *On vrach* ‘he (is a) doctor’, but past *On byl vrachom*.

Literature Review



Proceedings of the 1999 Conference of the Australian Linguistic Society 1

Towards a Cross-linguistic Typology of Copula Constructions

TIMOTHY JOWAN CURNOW
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- Four copula strategies: verbal, particle, inflectional and zero.
- Cross-linguistic variation in case marking.
- Constituent ordering.
- Grammatical cross-referencing.
- The difficulty of establishing universal grammatical relations.

Literature Review (Takings Incorporated in the Project)

1. Copula diversity across languages (Copula constructions do not follow one uniform pattern.)
Incorporated: Comparative analysis of five South Asian languages.

2. Zero copula as an important construction (Copular relationships can be expressed without an overt copula.)
Incorporated: Investigation of overt and zero-copula patterns.

3. Grammatical and semantic conditioning (Copula selection may depend on factors such as polarity and semantic relations.)
Incorporated: Analysis of predicate categories and polarity, non verbal).

4. Context-sensitive language variation (Copula constructions may vary according to discourse and grammatical context).
Extended in our study: Comparison of conversational and formal literary registers.

5. Need for empirical investigation (Typological descriptions provide a foundation, but actual language usage deserves examination).
Our extension: Corpus construction, computational extraction and quantitative analysis.

Research Gap

A quantitative question arising from the reviewed typology

Curnow (1999) provides a structural typology. Our study examines observed zero-copula rates in selected South Asian samples.

Dimension	Curnow (1999)	Our study
Evidence	Grammatical descriptions of about 70 languages	Counts from selected samples of five South Asian languages
Scope	Basic identity and classification constructions	Nominal, adjectival and locative/existential categories
Variation	Grammar and discourse condition construction choice	Compare rates by register and polarity within each language

Research question: How do zero-copula rates vary by register, predicate type and polarity in our five-language sample?

Reference: Curnow (1999), pp. 1-4. The gap is relative to the reviewed paper.

Our Study/ Methodology

Speech vs. writing

We built a cross-linguistic corpus of **non-verbal clauses** in five languages and measured how often the copula is missing in **casual spoken dialogue** versus **formal written prose**.

3 Five languages, two systems

Indo-Aryan · Hindi, Bengali, Assamese
An overt copula (hai, hoy) that speakers can leave out; Bengali uses none in the present.
vah mera bhai (hai) 'he is my brother'

Dravidian · Tamil, Telugu
No copula at all for 'X is Y'; location and negation always need an overt word.
avan en tambi 'he my brother'

4 What we asked

H1 · Register
Is the copula dropped more in casual film dialogue than in formal literary narration?

H2 · Predicate type
Is it dropped most with nouns ('he my brother'), less with adjectives, least with locations ('mother at home')?

H3 · Polarity
Do negative clauses (Hindi *vah taiyaar nahin*) drop the copula more often than affirmatives?

5 How we did it

01 Collect
Spoken: film subtitles, YouTube captions, informal dialogue. Formal: public-domain stories and essays (Premchand, Tagore, Gurajada …).

02 Extract
Python scripts clean the text, split sentences, separate literary dialogue from narration, and keep present-tense non-verbal clauses.

03 Tag
Each clause gets register, predicate type, polarity and copula status (1 = overt, 0 = zero);

04 Compare
Zero-copula rates per category, cross-tabulated within each language, then compared across the five languages.

Data Collection & Extraction

We built a cross-linguistic dataset across **5 South Asian languages**—Hindi, Bengali, Assamese (*Indo-Aryan*) and Tamil, Telugu (*Dravidian*)—comparing casual spoken dialogue against formal written prose.

01 *Movie Subtitles (Spoken)*

Extracted natural conversational dialogue from movie .srt subtitle files and word-timed YouTube auto-captions (json3), using Python scripts to strip timestamps, metadata credits, and song/music segments.

02 *Short Stories & Essays (Formal)*

Collected formal literary prose from public-domain .txt stories and essays (such as Premchand, Renu, Tagore, and Gurajada Apparao), splitting raw paragraphs into clean sentences using Indian script punctuation (।, ॥).

03 *Dialogue vs. Narrative Split*

Used Python regular expressions (re) to detect quotation marks ("...", «...»), dialogue dashes (—), and speech markers so character speech inside stories (**Literary Dialogue**) is separated from pure narration (**Literary Narrative**).

04 *Standardized Dataset Schema*

Unified all 5 languages into a consistent tabular CSV format tracking **Register**, **Sub-Register**, **Predicate Category**, **Polarity**, and **Copula Presence vs. Deletion**.

Isolating Non-Verbal Clauses in Python

How Python built-ins and NLP rules isolated pure present-tense non-verbal clauses before any counting was done.

01 *Python Built-In Filtering Pipeline*

- **Punctuation & Token Cleaning (string & re):** Used Python's built-in re and string.punctuation (extended with Indic dandas, ellipses, and em-dashes) to split clean word tokens, strip noise, and recover copulas attached to punctuation (like हैं—).
- **Tense & Action Verb Pruning (set.isdisjoint):** Converted tokenized sentences into Python set objects and used .isdisjoint() against curated word sets to instantly filter out past-tense markers (था/थी, गया), imperatives, and action verbs.
- **Suffix & Auxiliary Exclusion (str.endswith):** Applied Python tuple suffix checks (.endswith()) to remove future verb endings and present habitual/progressive action clauses (such as रहा है or जाता है) where the copula only acts as a helper verb.
- **Fragment Removal (len):** Filtered out 1–2 word exclamations and title headers to keep complete clauses.

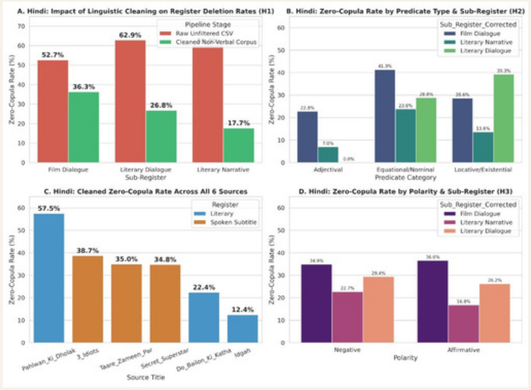
02 *Clause Tagging & Cleaning Impact*

- **Syntactic Role Tagging:** Combined Python lexical sets (spatial postpositions and negative words) with Stanza Universal Dependencies (root word and POS tags) to classify every valid non-verbal clause into:
  1. *Equational / Nominal* (noun/pronoun identity)
  2. *Adjectival* (qualities/states)
  3. *Locative / Existential* (location/possession)
- **Eliminating “False Zero” Errors:** Without Python's built-in verbal filters, normal past/action sentences lacking है get falsely counted as deleted copulas.
- **Cleaning Result:** Multi-stage Python auditing removed action-verb noise, correcting Hindi Literary Narrative copula deletion from an inflated **59.2% (raw)** down to an accurate **17.7% (clean)**.

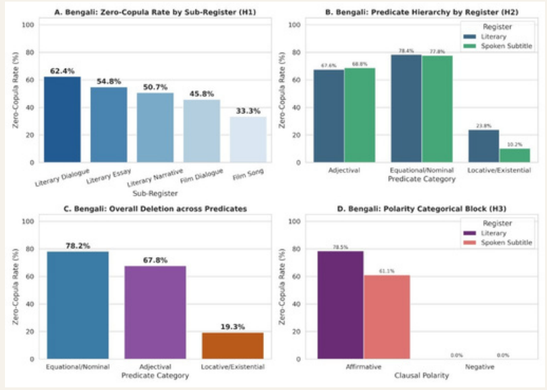
Cross-Linguistic Empirical Dashboards

All 5 Languages · Zero-Copula Rate (%)

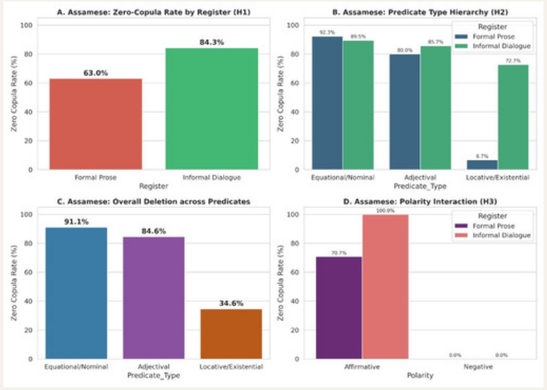
01 *Hindi (Central Indo-Aryan)*



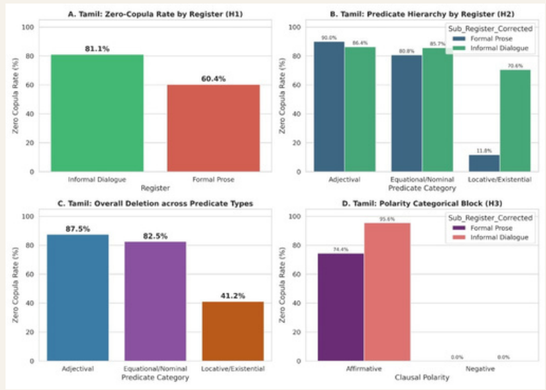
02 *Bengali (Eastern Indo-Aryan)*



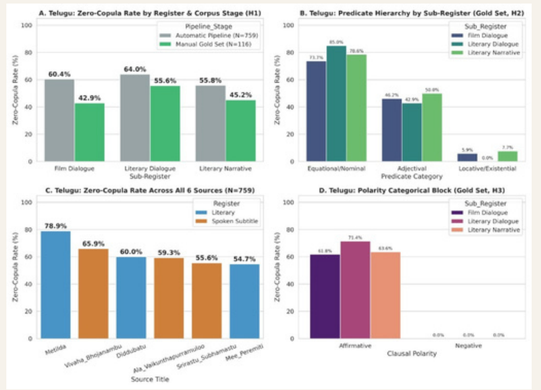
03 *Assamese (Eastern Indo-Aryan)*



04 *Tamil (South Dravidian)*



05 *Telugu (South-Central Dravidian)*



06 *Key Cross-Linguistic Patterns*

- **Register Difference:** Hindi drops the copula much more in casual film speech (**36.3%**) than formal narration (**17.7%**). In Bengali, Assamese, Tamil, and Telugu, zero-copula in affirmative identity sentences is standard grammar across registers.
- **Predicate Hierarchy:** In all 5 languages, *Equational/Nominal* and *Adjectival* sentences show high copula deletion, whereas *Locative/Existential* sentences strongly keep overt verbs (retaining the verb ~60%–95% of the time).
- **Negation Rule:** Bengali, Assamese, Tamil, and Telugu show **0.0% deletion** in negative clauses, while Hindi freely allows deletion (**34.9%**).

Comparison with Published Reports & Nuances after Analysis.

How our cross-linguistic dashboards line up with typological literature—and where genre matters.

01 *Alignment with Linguistic Literature*

- **Universal Locative Retention:** Confirms cross-linguistic typology (Curnow 1999; Stassen 1997)—expressing physical location or existence requires an explicit existential verb (*hai, aachhe, ase, irukku, undi*) so meaning and time reference are not lost.
- **Why Negation Blocks Deletion:** In Hindi, negation uses a standalone word (*nahin*) that carries emphasis on its own, allowing *hai* to drop (Kachru 2006). In Bengali, Assamese, Tamil, and Telugu, negation is fused into dedicated negative verbs (*nei, nai, illai, kadu/ledu*)—dropping the verb would erase the negative meaning itself.

02 *Dataset & Genre Observations*

- **Natural Text vs. Paired Sentence Lists:** Continuous film and story corpora (Hindi, Bengali, Telugu) measure how often speakers actually drop the copula in running text, while paired acceptability sets (Tamil, Assamese) also capture short conversational replies (*"Where are you?"* → *"At home"*) where spoken locatives can omit the verb.
- **Story Style Variation:** Regionalist fiction featuring folk rhythms and rural speech (*Pahlwan Ki Dholak*, **57.5%** deletion) behaves like spoken drama compared to standard narrative prose (*Idgah*, **12.4%** deletion).

Register Differences Depend on Predicate Type

Locative/existential predicates show the largest register gaps in the Tamil and Assamese samples

01 Evidence from Our Dashboards

Locative/existential zero-copula rate (%)

Language	Formal	Informal	Gap (pp)
Tamil	11.8%	70.6%	**58.8**
Assamese	6.7%	72.7%	**66.0**

Nominal and adjectival gaps are much smaller: 2.8 to 5.7 percentage points.

Formal = formal prose. Informal = informal dialogue.
pp = percentage points.

02 What Our Comparison Adds

A pattern beyond the overall rate

The register association varies by predicate type. Overall percentages conceal the much larger differences in locative/existential clauses.

Relation to Curnow (1999)

Curnow studies basic identity and classification constructions. Our analysis compares observed zero rates across registers and predicate categories.

These descriptive differences support further testing. They do not establish a universal rule or statistical significance.

Limitations & Key Findings

Scope of the study and the findings our comparison contributes

01 Limitations

Limited language coverage

Five languages from two families cannot represent the full linguistic diversity of South Asia.

Limited coverage within languages

Dialects and regional or social varieties lack systematic coverage.

Source and sample differences

Selected sources and unequal samples limit direct comparisons and coverage of everyday speech.

Annotation uncertainty

Filtering and annotation may retain errors. Larger manually checked samples are needed.

02 Key Findings & Contribution

Register differences vary by predicate type

Locative/existential gaps reach 58.8 percentage points in Tamil and 66.0 in Assamese.

Predicate rankings differ across samples

Bengali: nominal 78.2%, adjectival 67.8%.
Tamil: nominal 82.5%, adjectival 87.5%.

A basis for further comparative work

The comparison identifies specific patterns to test in larger corpora with better matched sources.

Main takeaway **Our findings show why zero-copula comparisons should consider predicate type alongside register.**

Conclusion & Future Work

What the five-language comparison shows for our three hypotheses

01 Hypotheses vs. Evidence

H1 · Register: partly supported

Hindi drops the copula in 36.3% of casual film clauses vs 17.7% of formal narration. Elsewhere zero is grammar in both registers; Tamil and Assamese gaps are mainly in locatives (58.8 and 66.0 pp).

H2 · Predicate type: supported

Locatives keep an overt verb in all five languages (about 60%–95%). Nominal vs adjectival order differs: Bengali 78.2% vs 67.8%, Tamil 82.5% vs 87.5%.

H3 · Polarity: not supported outside Hindi

Negatives show 0.0% deletion in Bengali, Assamese, Tamil and Telugu. Hindi allows deletion (34.9%).

02 Conclusions & Future Work

Grammar before style

In Bengali, Assamese, Tamil and Telugu, zero copula in identity clauses is standard grammar in both registers, and negation is fused into a negative verb (nei, nai, illai, kadu/ledu).

Hindi is the exception

Its overt copula (hai) is optional, so register and negation can license deletion.

Future work

- Larger, manually checked samples to test significance
- Matched sources and text types across languages
- More languages, dialects and social varieties

Predicate type shapes zero copula in all five languages; register and polarity matter mainly in Hindi, plus locatives in Tamil and Assamese.

Thank You